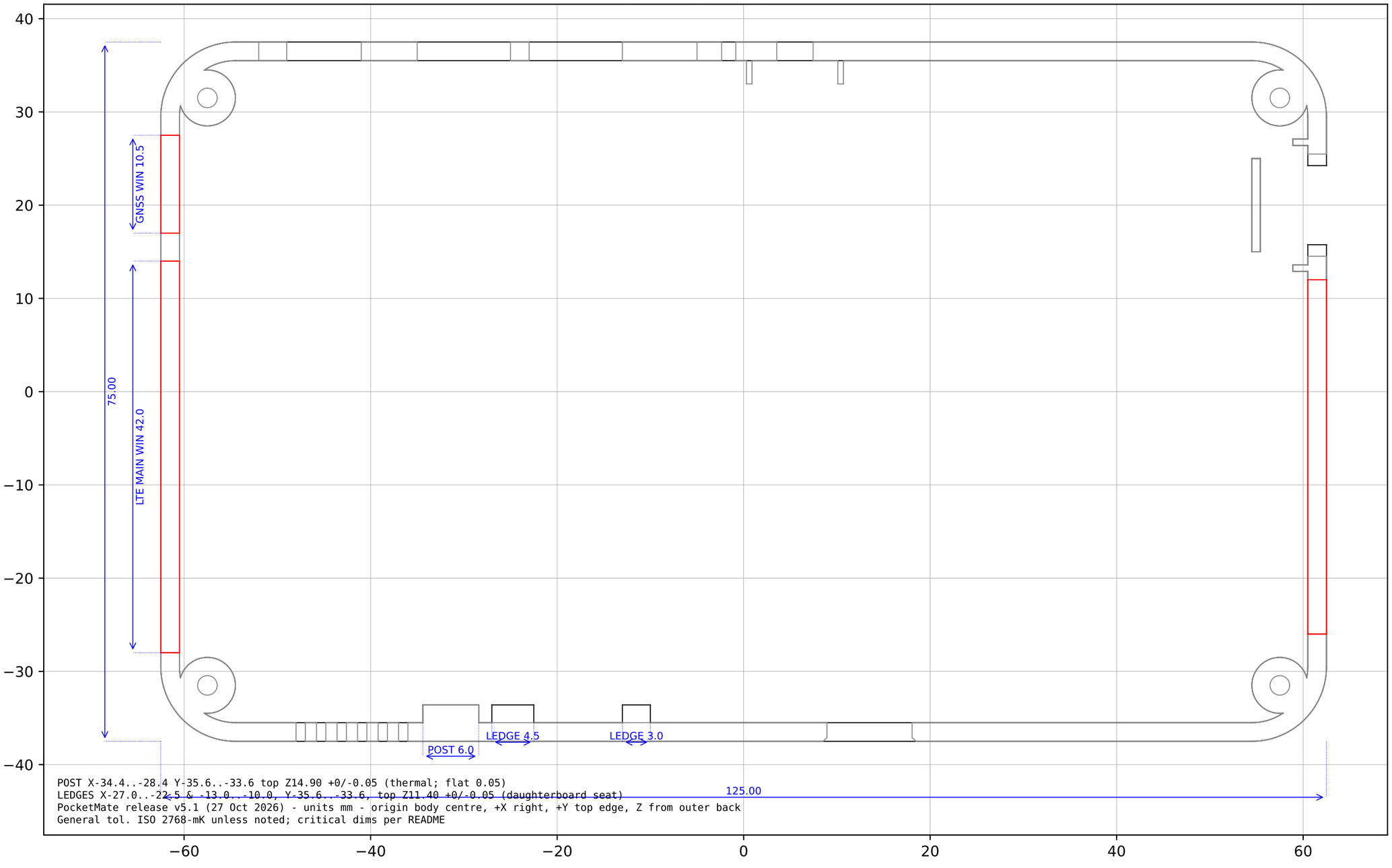
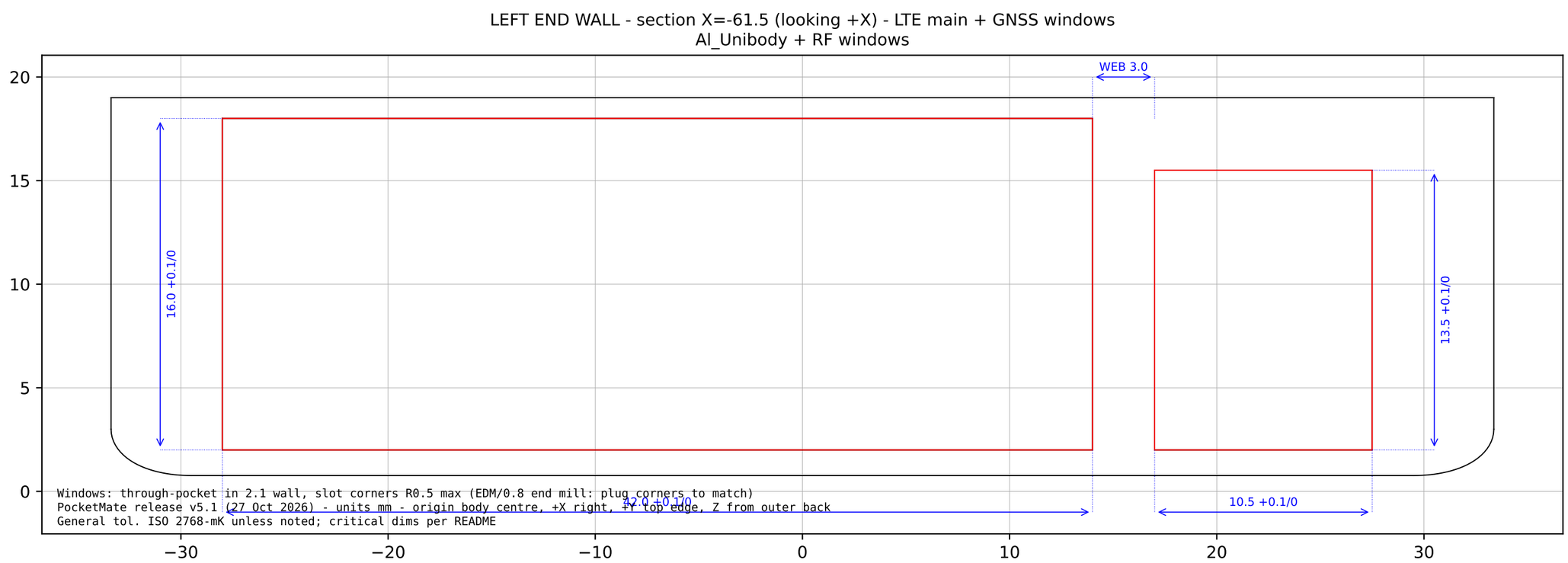
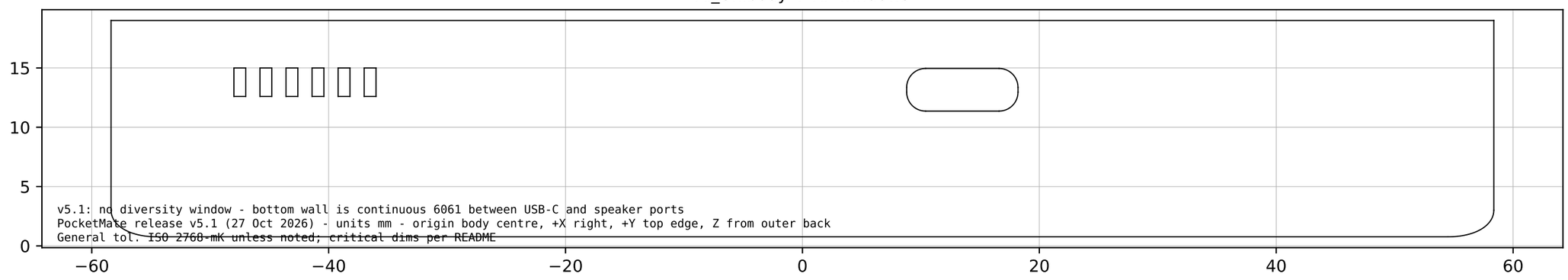
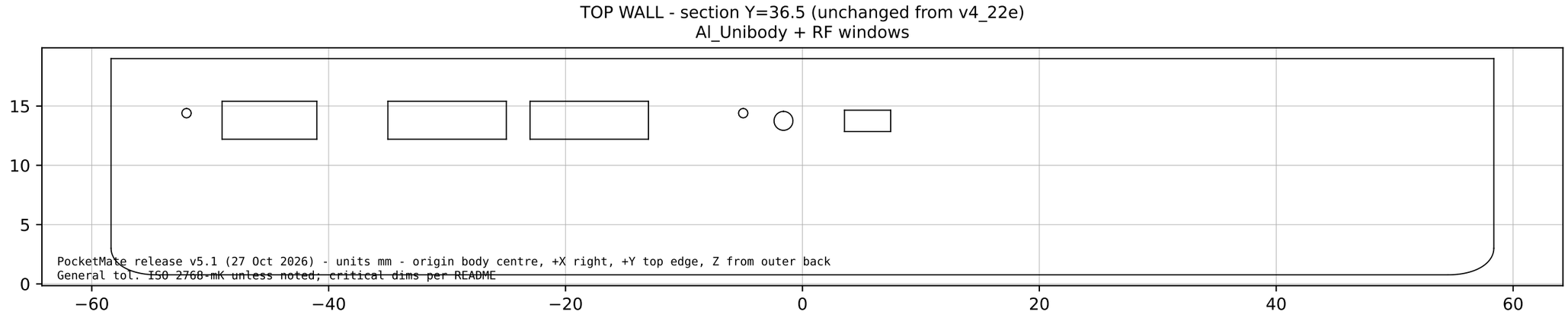
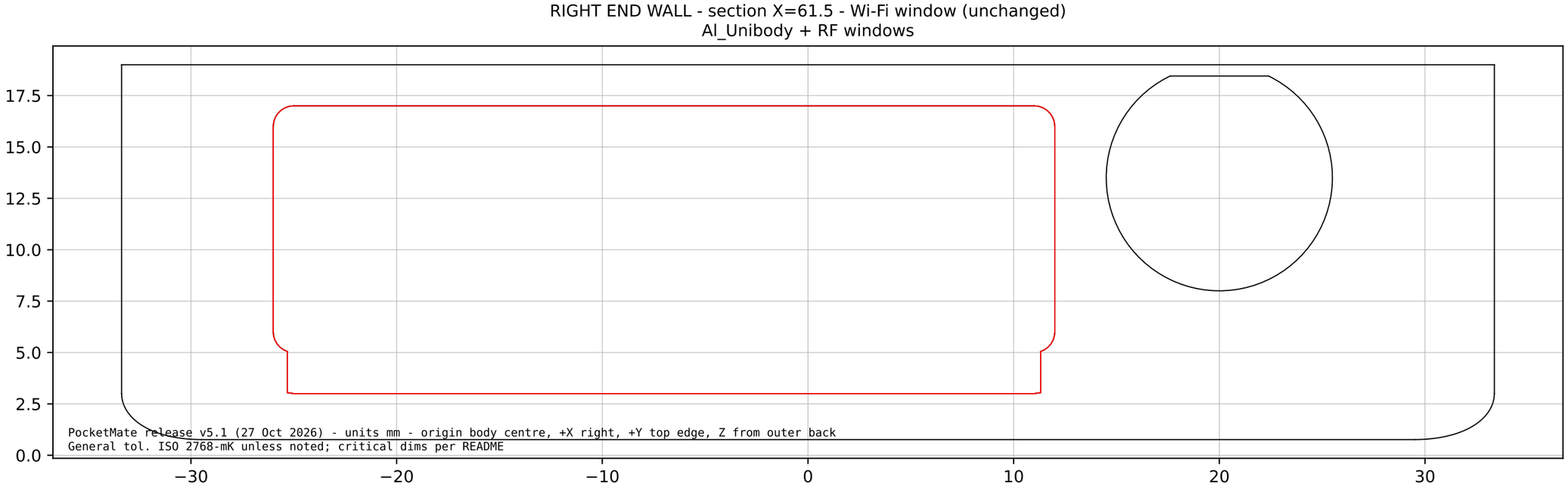
UNIBODY 6061-T6 - plan section Z=10.0 (cell/CM5 level) and Z=14.0 (LTE shelf level)
Al_Unibody_release_v5.step



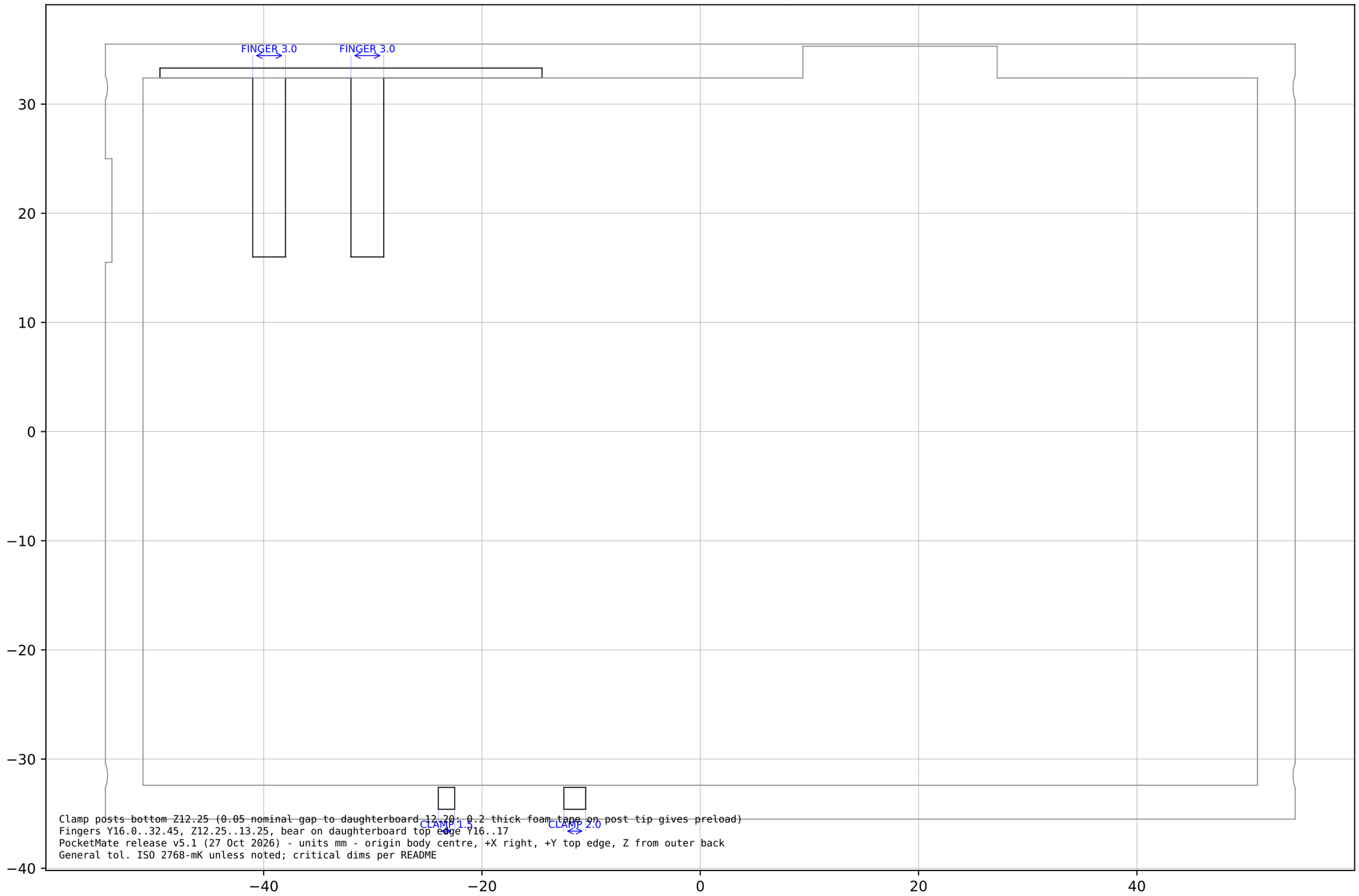

LEFT END WALL - section X=-61.5 (looking +X) - LTE main + GNSS windows
Al_Unibody + RF windows
WEB 3.0
16.0 +0.1/0
13.5 +0.1/0
42.0 +0.1/0
10.5 +0.1/0
Windows: through-pocket in 2.1 wall, slot corners R0.5 max (EDM/0.8 end mill: plug corners to match)
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README
20
15
10
5
0
−30
−20
−10
0
10
20
30


BOTTOM WALL - section Y=-36.5 - plain Al (v5.1: diversity window deleted)
Al_Unibody + RF windows
15
10
5
0
−60
−40
−20
0
20
40
60
v5.1: no diversity window - bottom wall is continuous 6061 between USB-C and speaker ports
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted, critical dims per README


TOP WALL - section Y=36.5 (unchanged from v4_22e)
Al_Unibody + RF windows
15
10
5
0
−60
−40
−20
0
20
40
60
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted, critical dims per README


RIGHT END WALL - section X=61.5 - Wi-Fi window (unchanged)
Al_Unibody + RF windows
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−30
−20
−10
0
10
20
30
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

BEZEL / DISPLAY FRAME (PC) - plan sections Z=13.0 (clamps, fingers) and Z=16.4 (lips)
Bezel_DisplayFrame_PC_release_v5.step

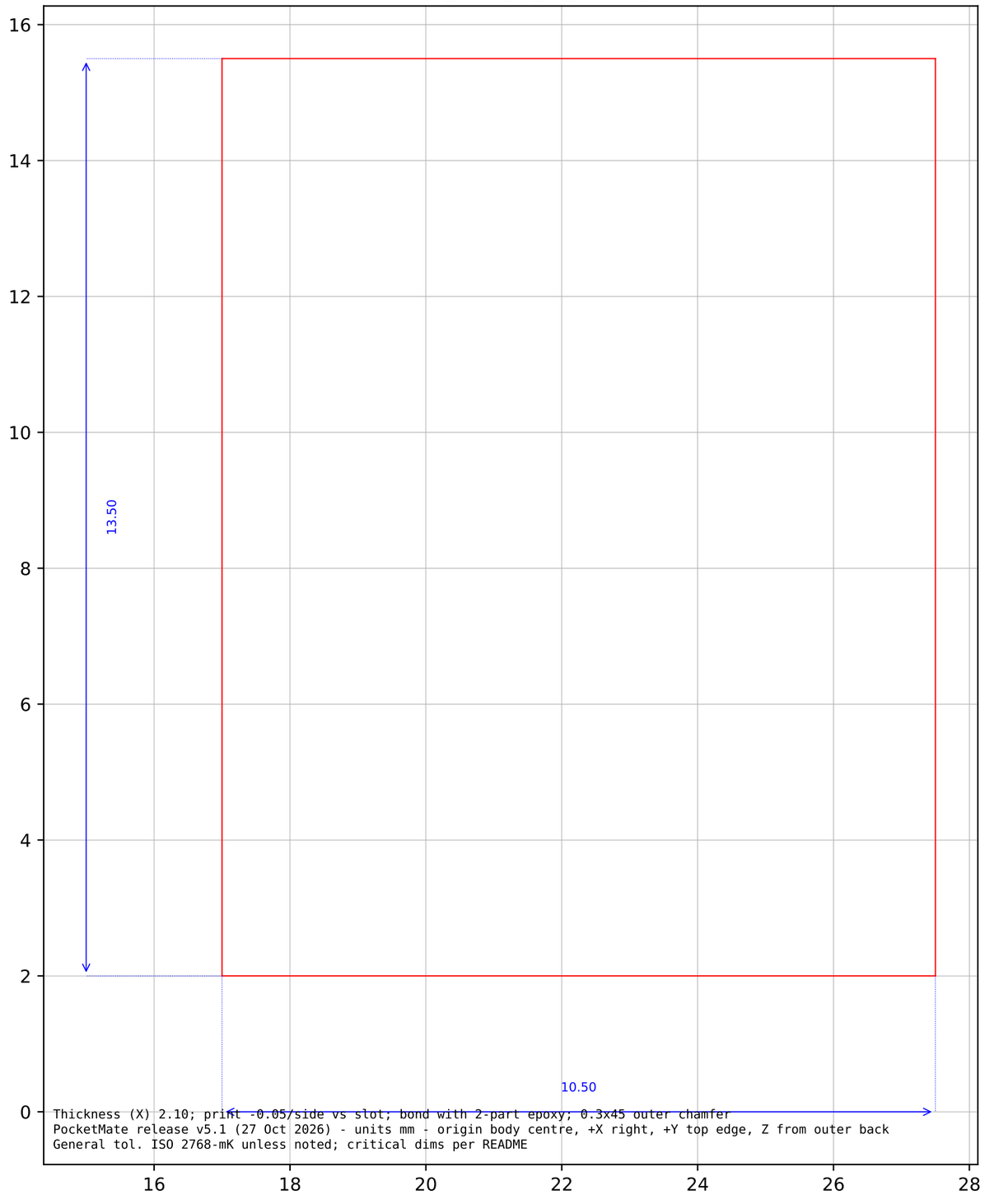


Clamp posts bottom Z12.25 (0.05 nominal gap to daughterboard 12.20; 0.2 thick foam tape on post tip gives preload)
Fingers Y16.0..32.45, Z12.25..13.25, bear on daughterboard top edge Y16..17
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

RF WINDOW RFWindow_LTE_Main_L_PC - SLA clear/tough resin (NO metal-filled resin)
stl/RFWindow_LTE_Main_L_PC_release_v5.stl

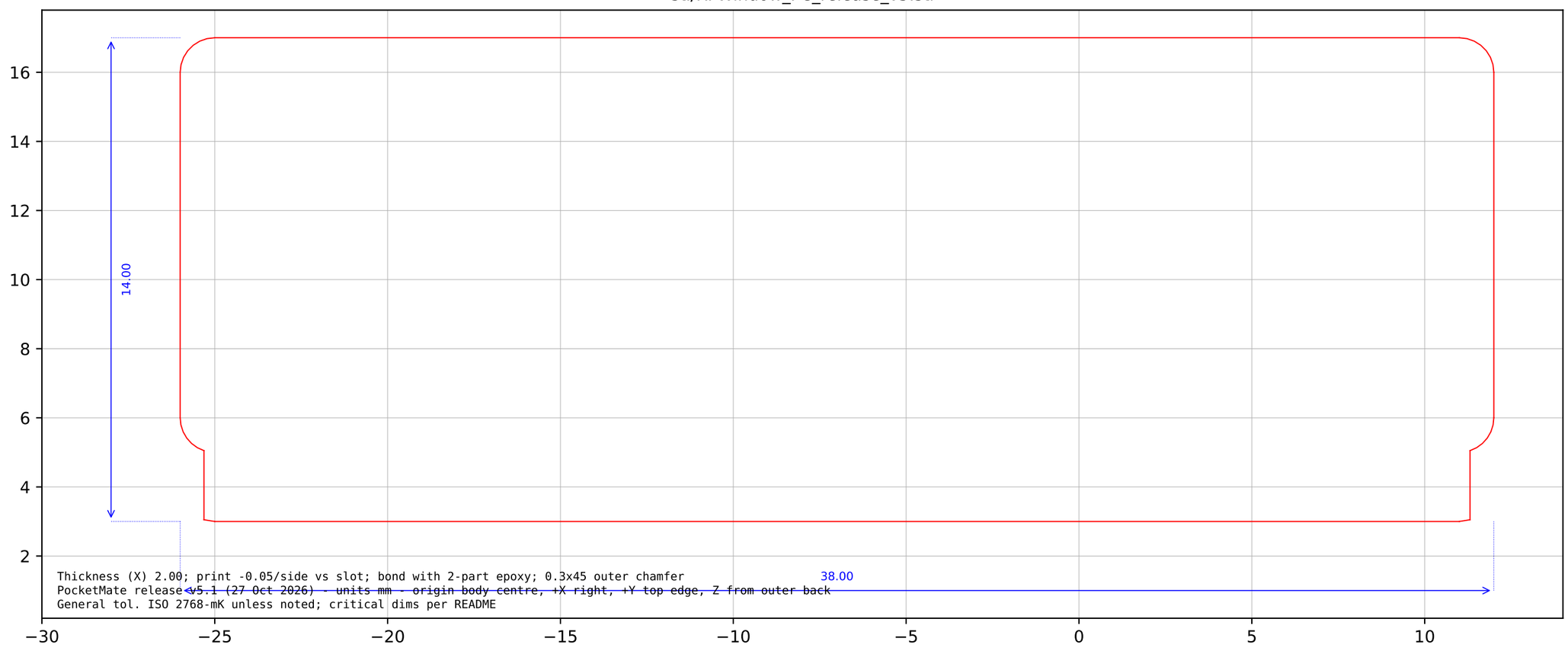


RF WINDOW RFWindow_GNSS_TL_PC - SLA clear/tough resin (NO metal-filled resin)
stl/RFWindow_GNSS_TL_PC_release_v5.stl



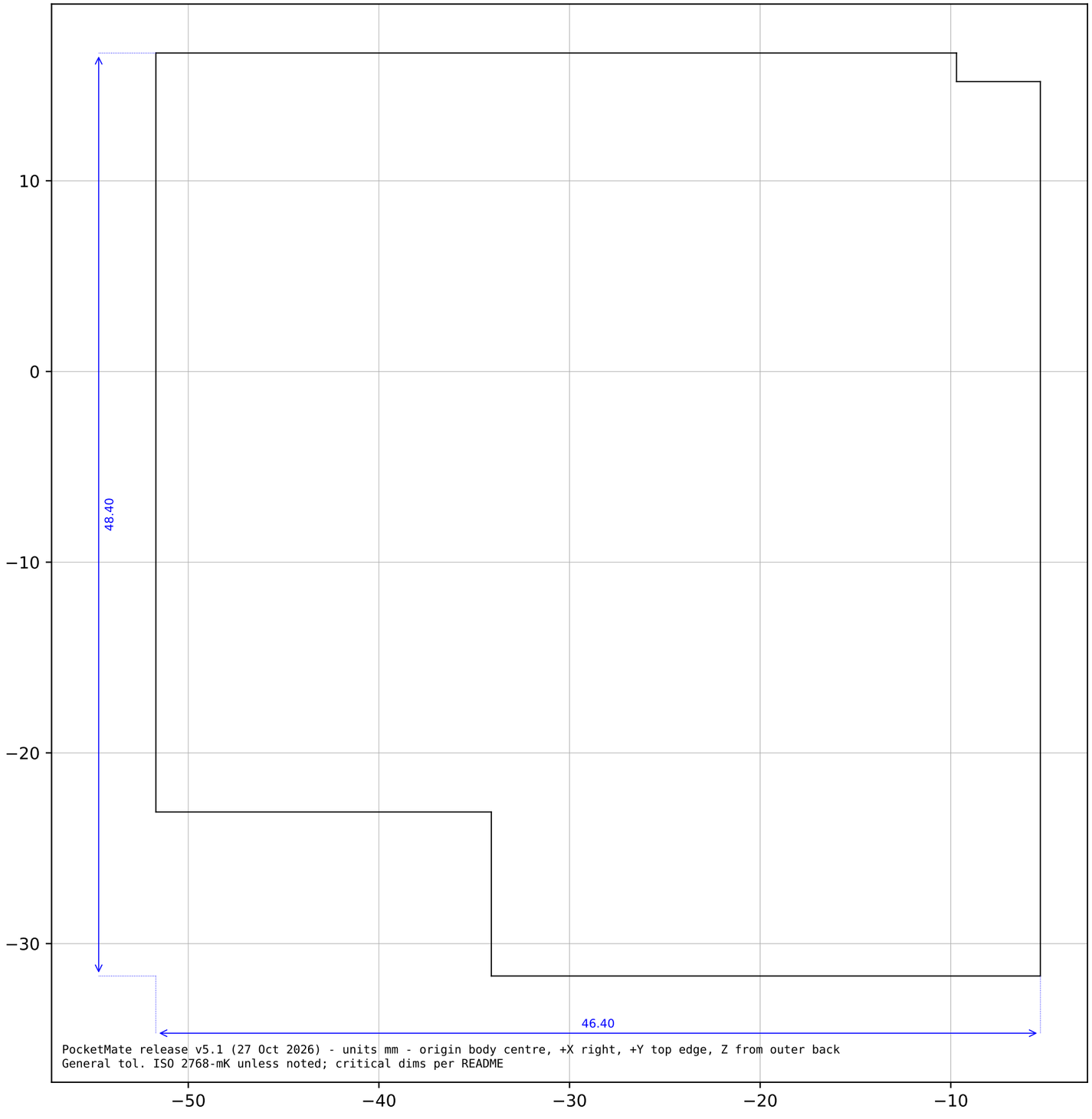
Thickness (X) 2.10; print -0.05/side vs slot; bond with 2-part epoxy; 0.3x45 outer chamfer
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

RF WINDOW RFWindow_PC - SLA clear/tough resin (NO metal-filled resin)
stl/RFWindow_PC_release_v5.stl



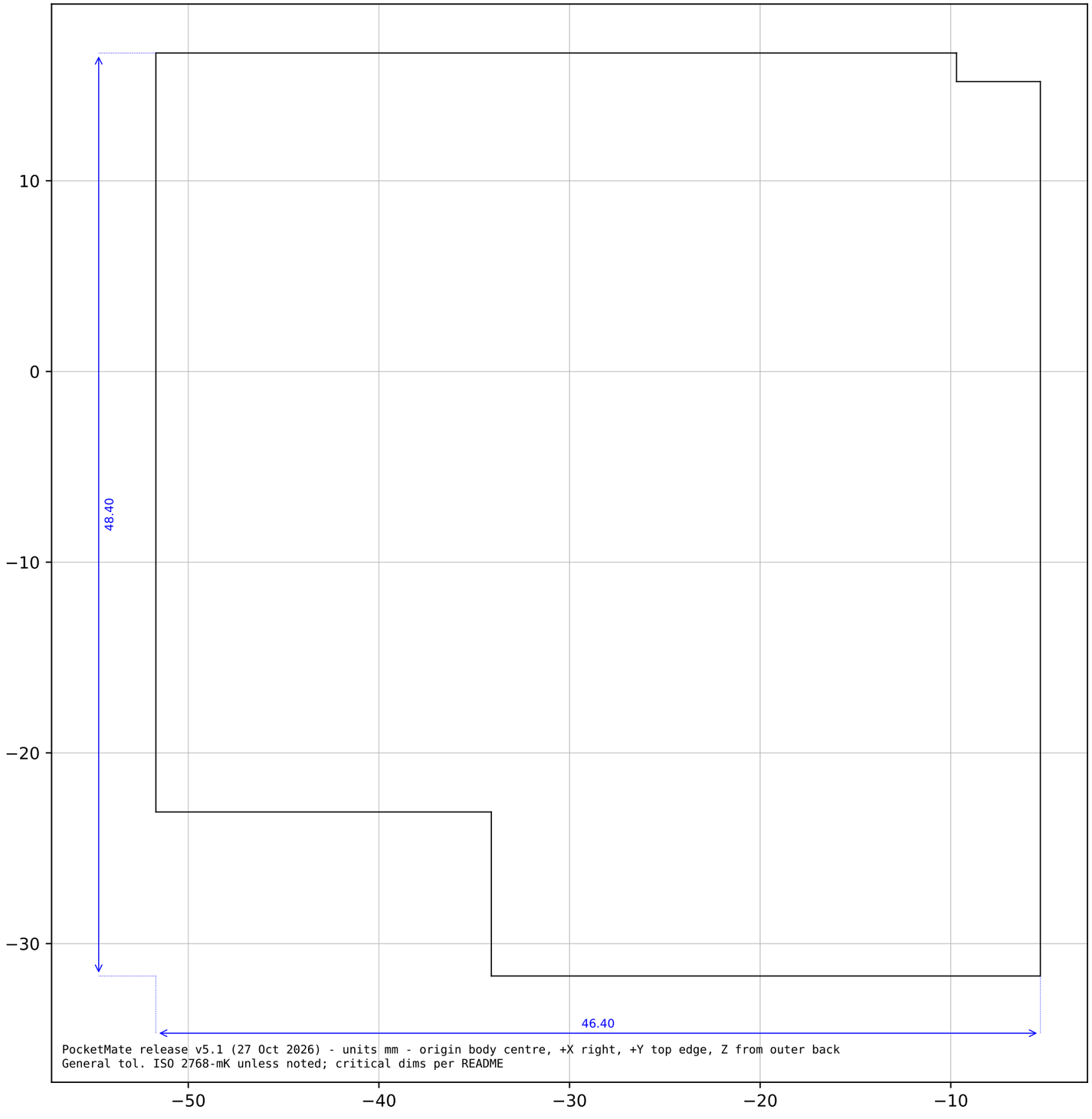
Thickness (X) 2.00; print -0.05/side vs slot; bond with 2-part epoxy; 0.3x45 outer chamfer
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

FLAT PATTERN Foam_Prod_0p5
PE/PU closed-cell foam 0.5 (production cell) - die/laser cut



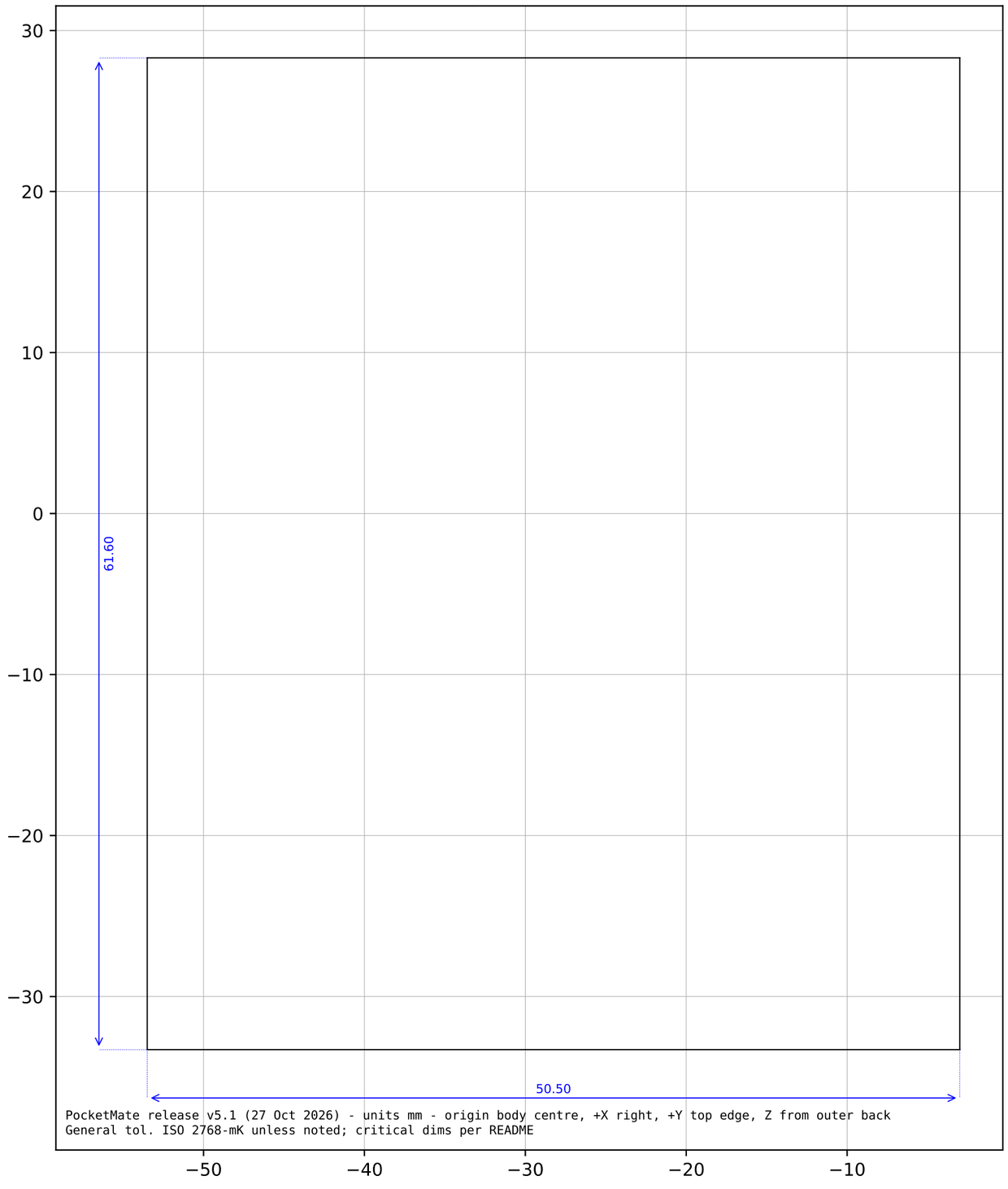
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

FLAT PATTERN Foam_Proto_2p0
PE/PU closed-cell foam 2.0 (Akyga proto cell)

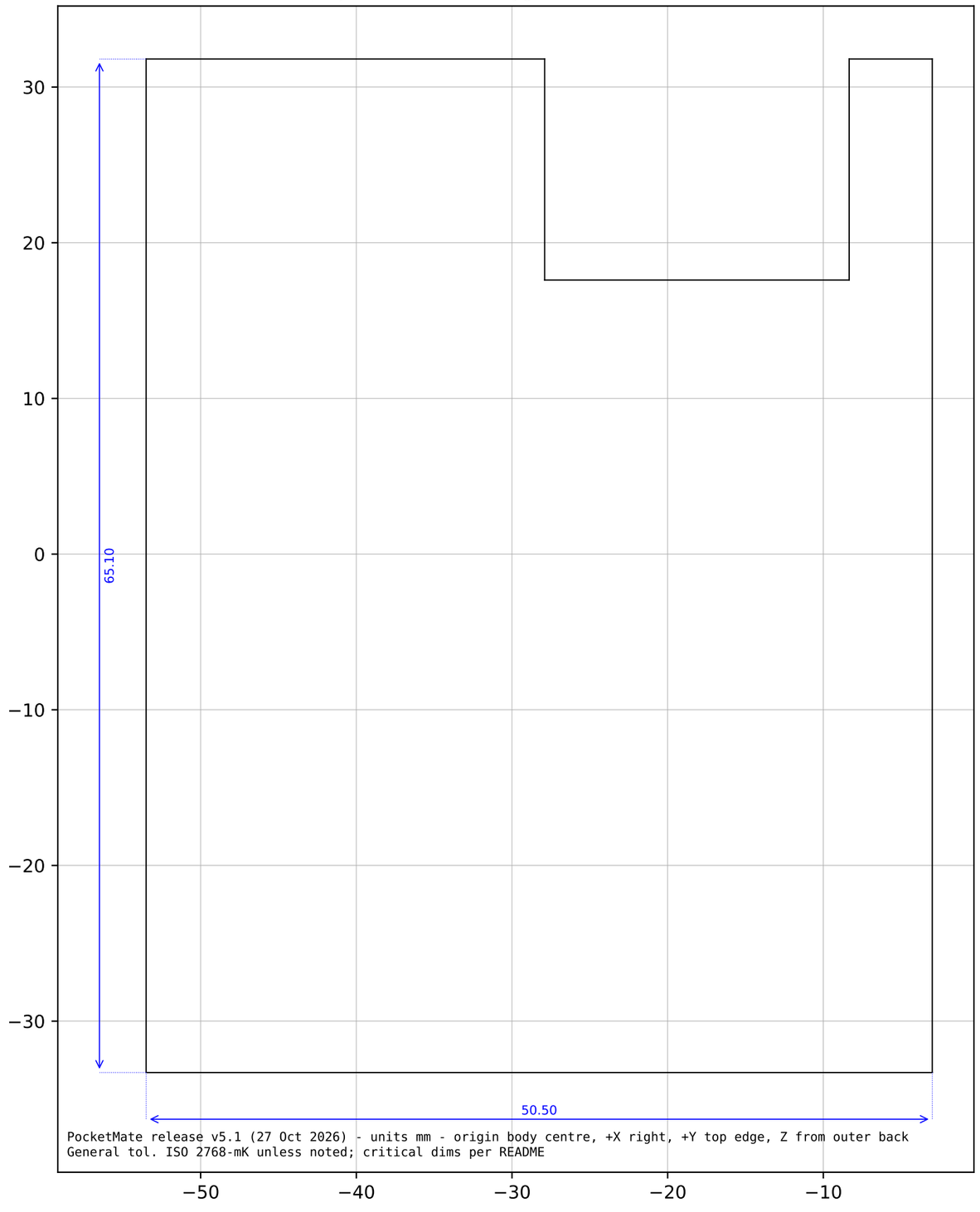


PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

FLAT PATTERN Liner_PET_Prod
PET/polyimide 0.1 insulating liner, adhesive one side, onto cell top

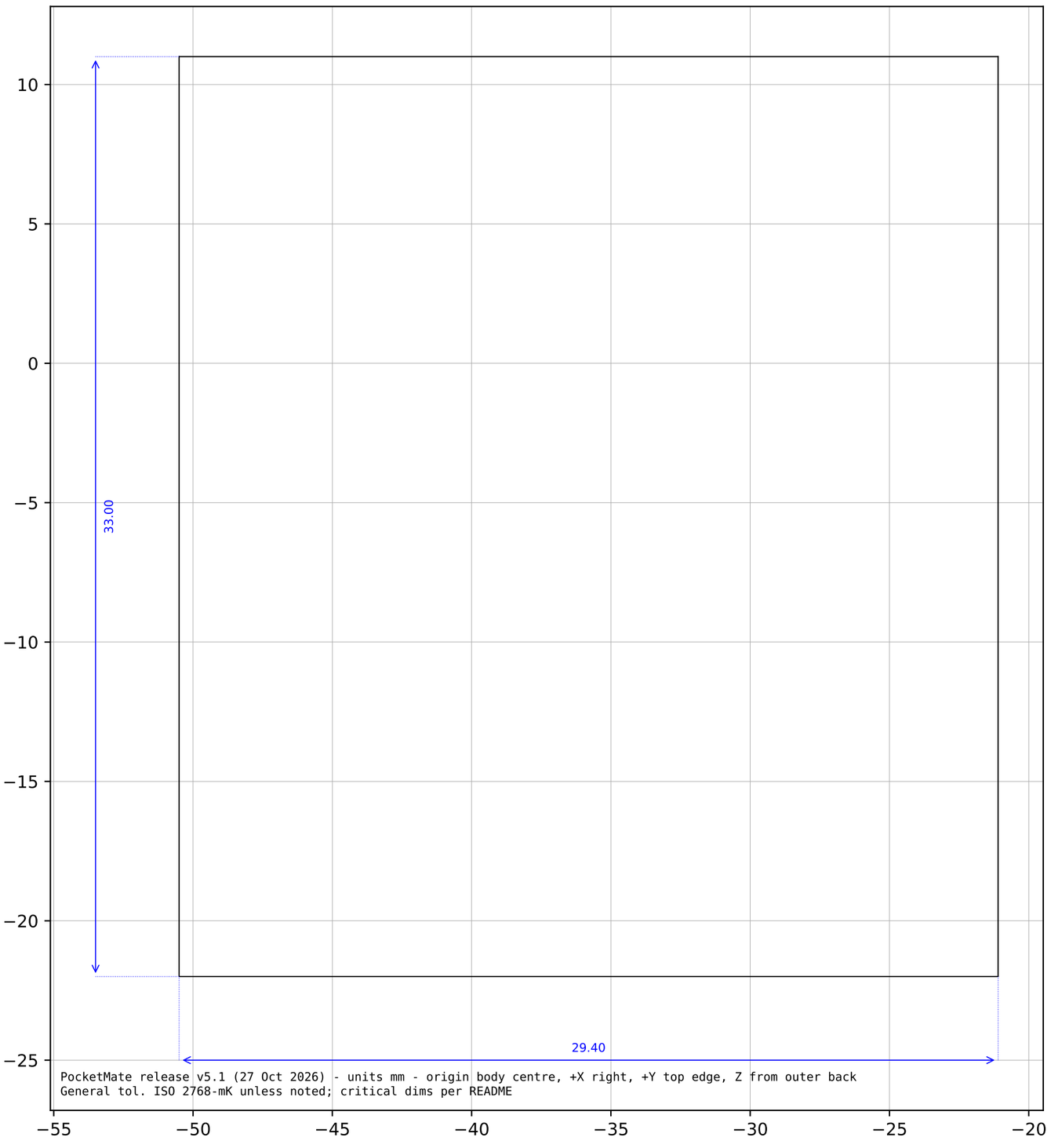


FLAT PATTERN Liner_PET_Proto
PET/polyimide 0.1 liner, Akyga cell; relief around leads

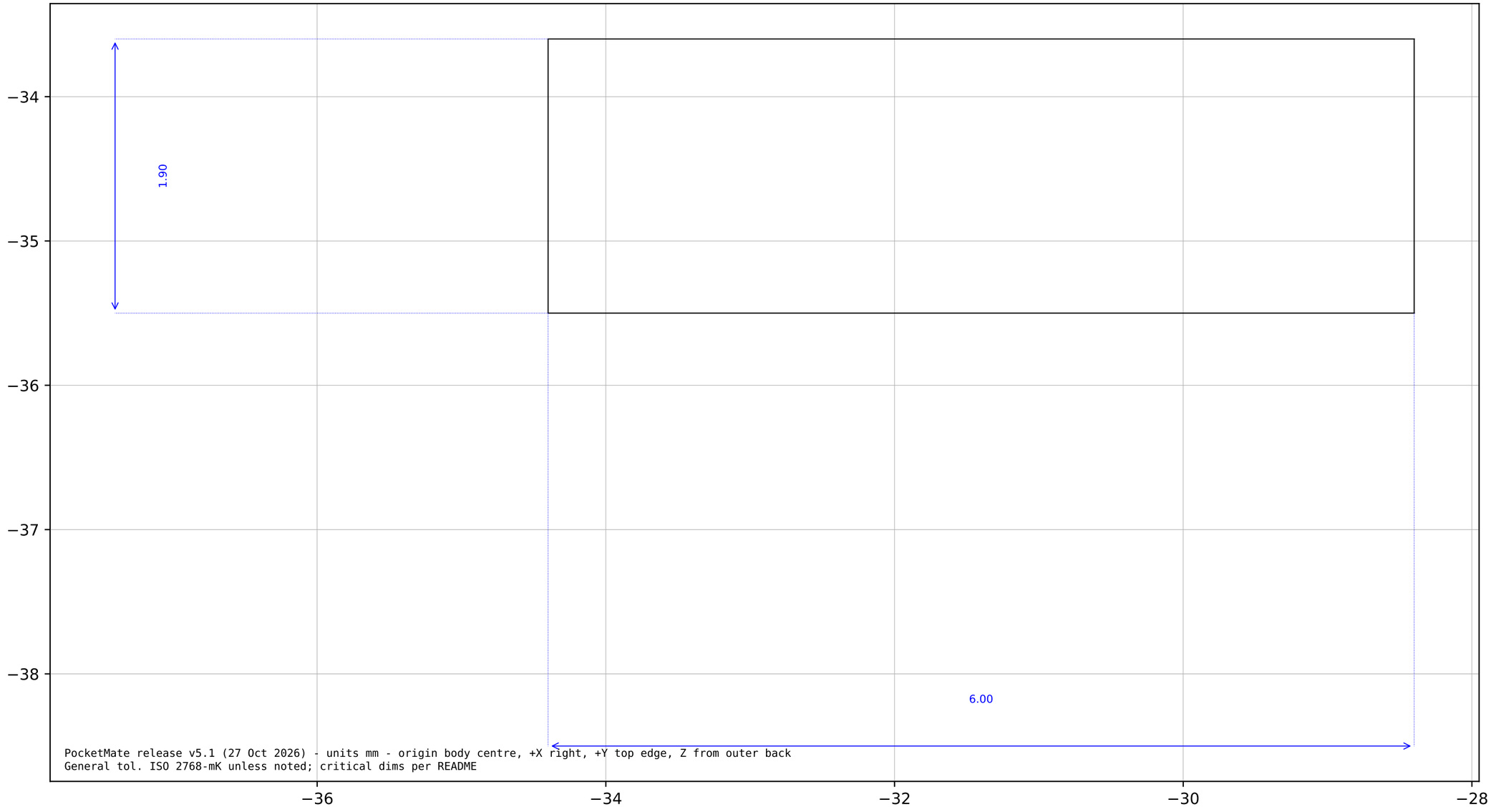


PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

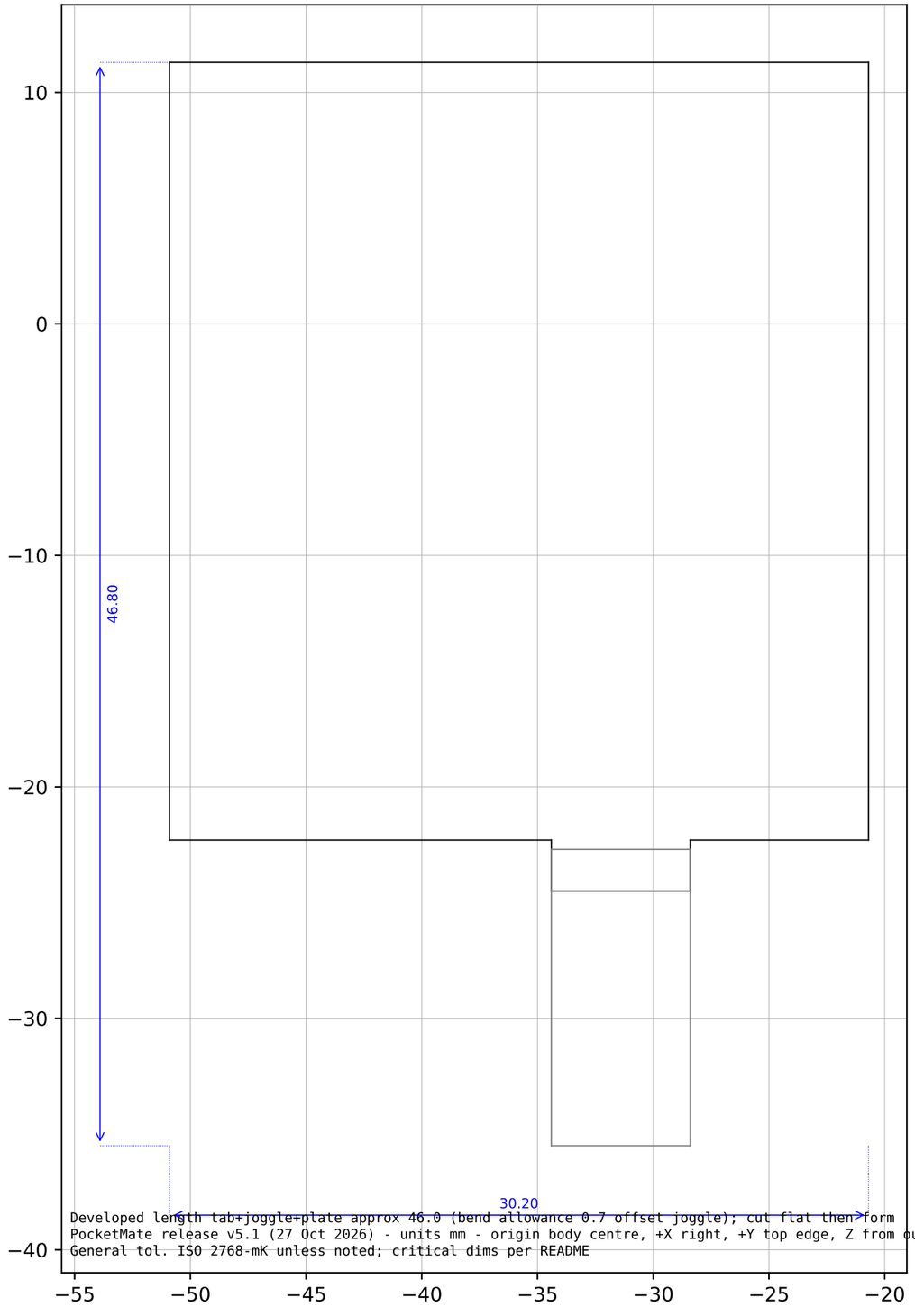
FLAT PATTERN ThermalPad_Shield_0p5
Gap pad 0.5, >=3 W/mK, 29.4 x 33.0



FLAT PATTERN ThermalPad_Post_0p3
Gap pad 0.3, >=3 W/mK, 6.0 x 1.9

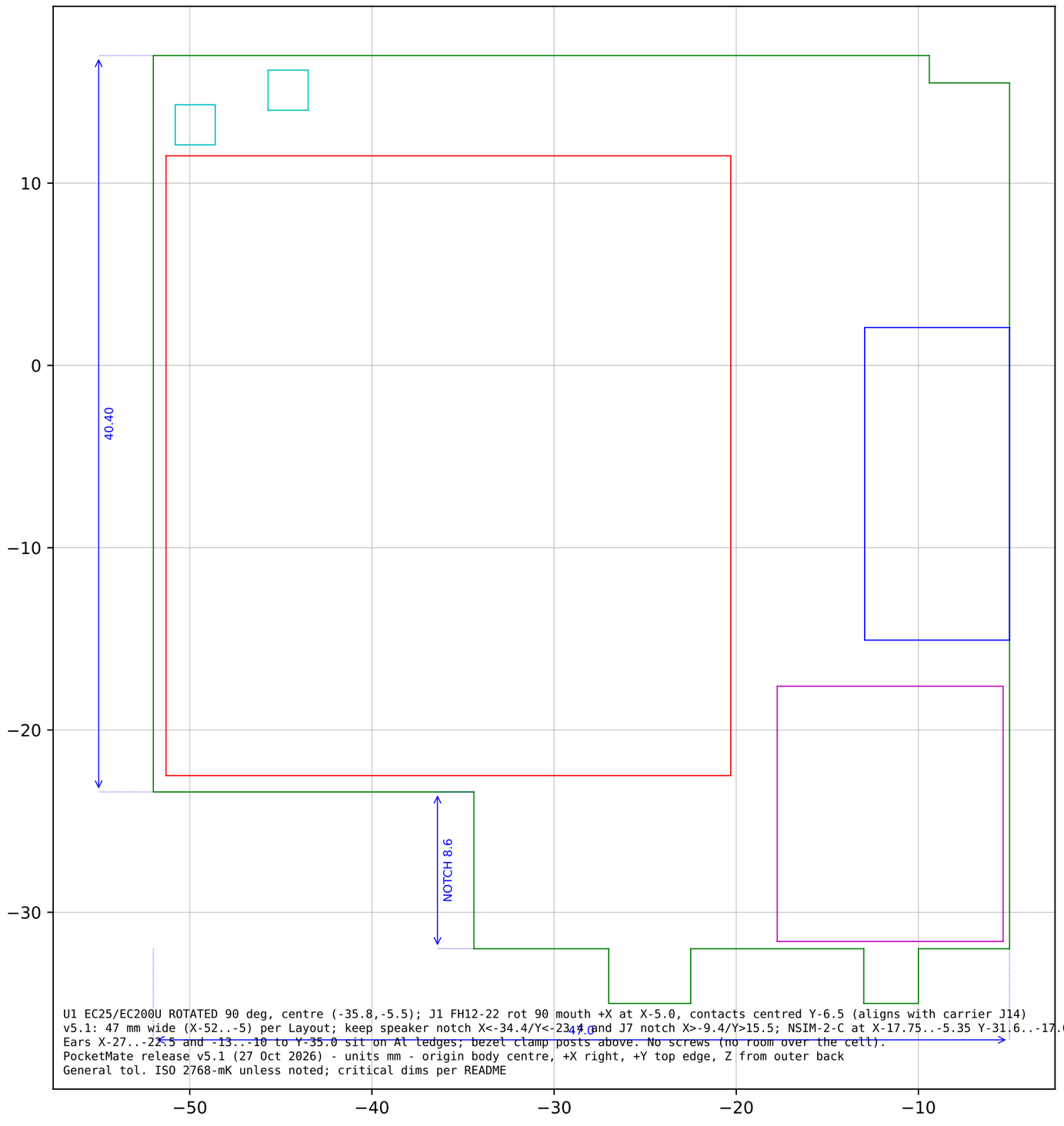


FLAT PATTERN HeatSpreader_Cu_0p5
C1100 Cu 0.5, Ni plate; joggle 0.7 down at Y-22.7/-24.5; Kapton 0.05 on top face (display side)



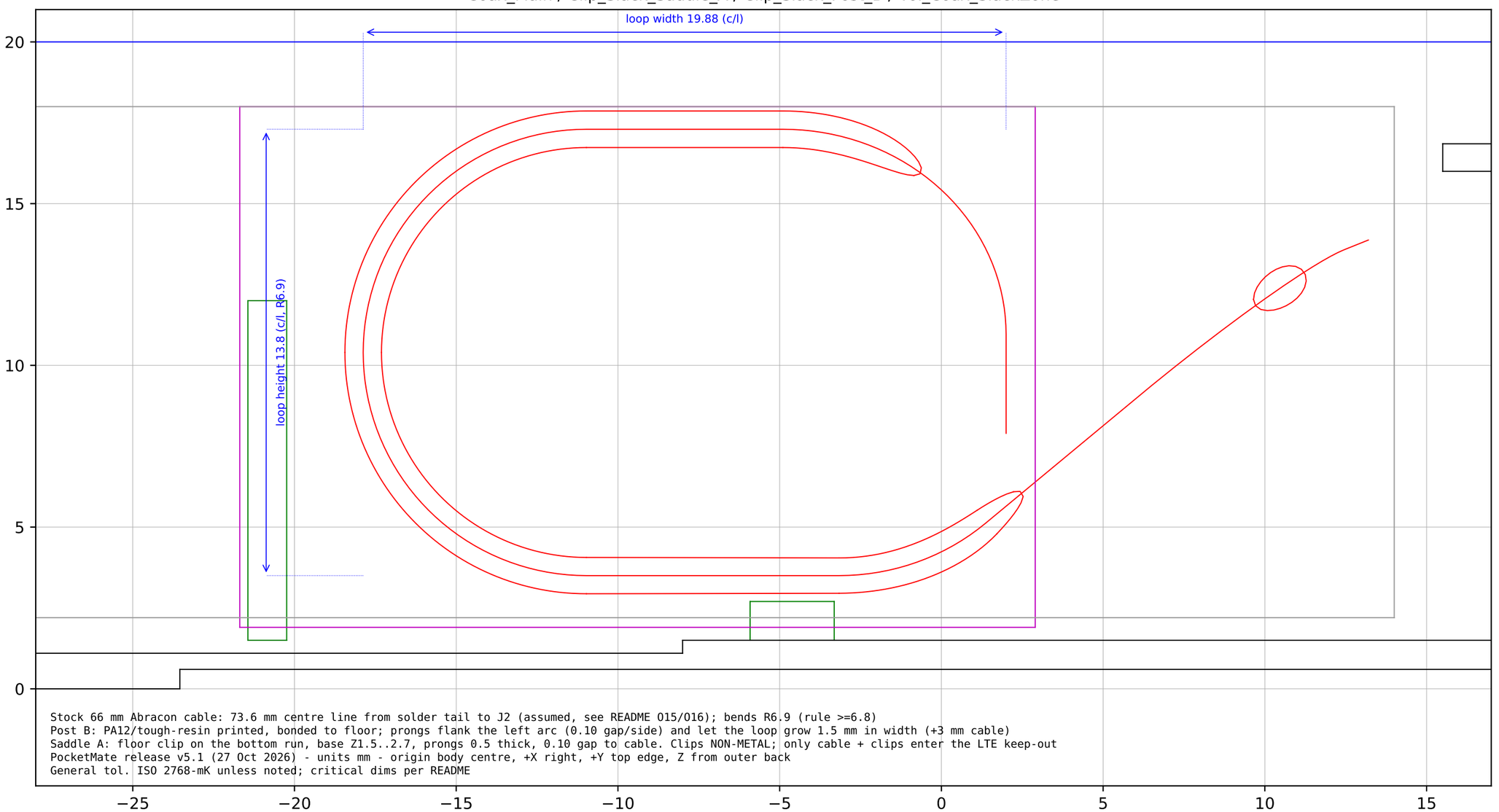
Developed length tab+joggle+plate approx 46.0 (bend allowance 0.7 offset joggle); cut flat then form
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

LTE DAUGHTERBOARD OUTLINE (Edge.Cuts proposal for hardware/kicad-lte) - 0.8 mm 4L
mech coords; KiCad Y = -mech Y



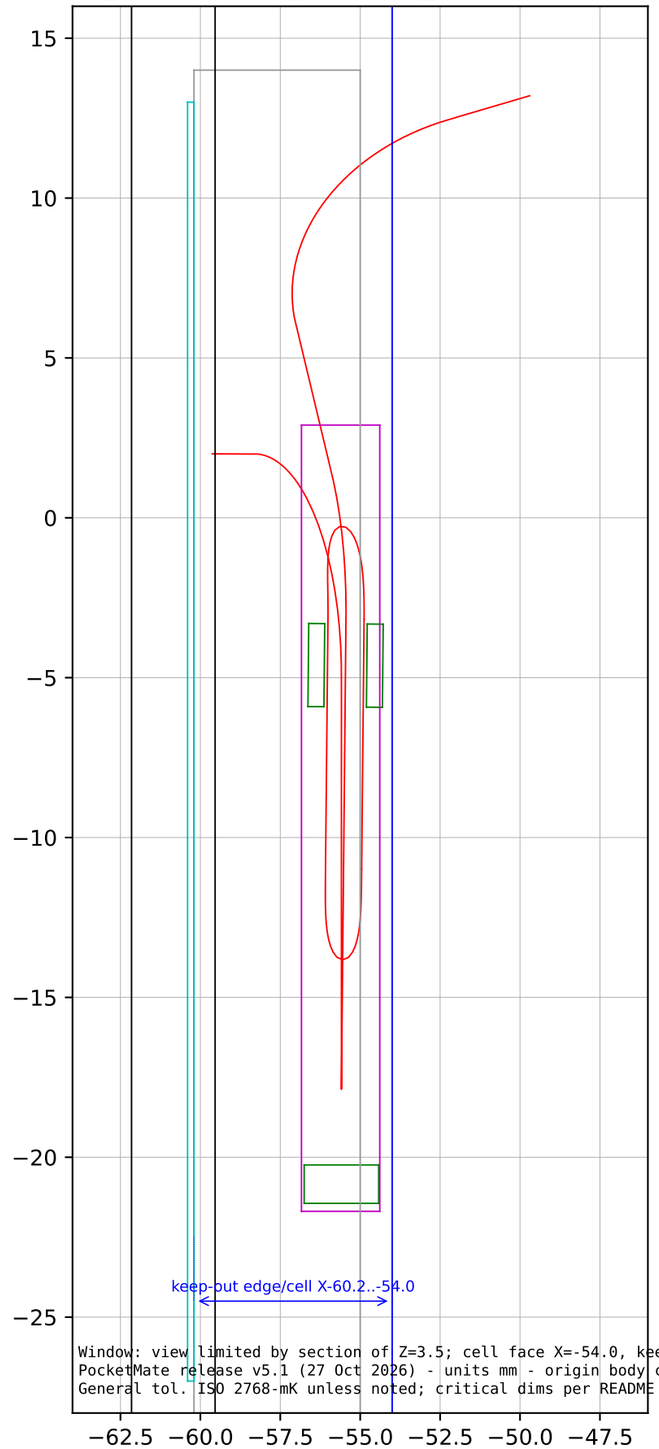
U1 EC25/EC200U ROTATED 90 deg, centre (-35.8,-5.5); J1 FH12-22 rot 90 mouth +X at X-5.0, contacts centred Y-6.5 (aligns with carrier J14)
v5.1: 47 mm wide (X-52..-5) per Layout; keep speaker notch X<-34.4/Y<-23.4 and J7 notch X>-9.4/Y>15.5; NSIM-2-C at X-17.75..-5.35 Y-31.6..-17.6
Ears X-27..-22.5 and -13..-10 to Y-35.0 sit on Al ledges; bezel clamp posts above. No screws (no room over the cell).
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

O15 STOCK-CABLE SLACK LOOP - elevation, section plane X=-55.59 (view from +X), Coax_Main centre line + clips + zone
Coax_Main / Clip_Slack_Saddle_A / Clip_Slack_Post_B / Vol_Coax_SlackZone



Stock 66 mm Abracon cable: 73.6 mm centre line from solder tail to J2 (assumed, see README O15/O16); bends R6.9 (rule >=6.8)
Post B: PA12/tough-resin printed, bonded to floor; prongs flank the left arc (0.10 gap/side) and let the loop grow 1.5 mm in width (+3 mm cable)
Saddle A: floor clip on the bottom run, base Z1.5..2.7, prongs 0.5 thick, 0.10 gap to cable. Clips NON-METAL; only cable + clips enter the LTE keep-out
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README

O15 STOCK-CABLE SLACK LOOP - plan section Z=3.5 (bottom run, clips, cell face)
Coax_Main / clips / Vol_Battery / Al_Unibody / FPC volume



Window: view limited by section of Z=3.5; cell face X=-54.0, keep-out X-60.2..-55.0, FPC X-60.4..-60.2; floor top Z1.5
PocketMate release v5.1 (27 Oct 2026) - units mm - origin body centre, +X right, +Y top edge, Z from outer back
General tol. ISO 2768-mK unless noted; critical dims per README